

Building AI Agents with N8N

Introduction

This guide serves as a hands-on training material to help you create your own AI Agents in n8n from scratch. We will cover the entire process, including creating the workflow, adding the necessary nodes, connecting your first chat model, and integrating a Check Point MCP server (`gw-cli`).

Overview

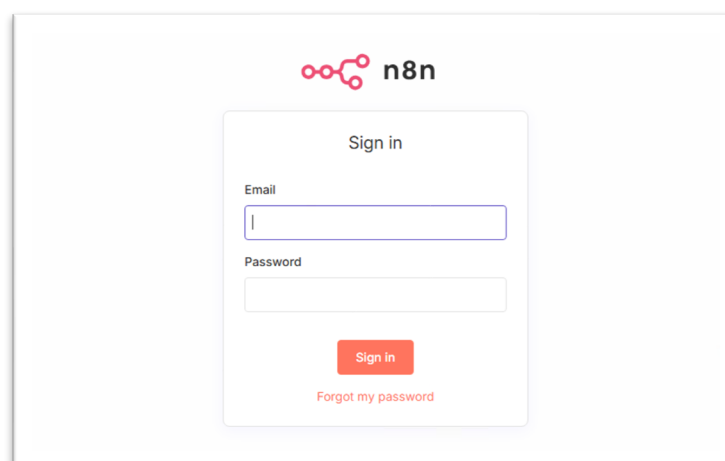
The Quantum Gateway CLI MCP server provides a conversational interface to the Check Point Gateway CLI (Clish/Expert mode commands, depending on implementation). It simplifies the execution of complex CLI commands while running them centrally through the Management API.

Prerequisites (it is pre-configured in this lab)

- **n8n URL:** <http://10.1.3.33:5678>
- **Credentials:**
 - **OpenAI / Azure OpenAI / Ollama:** For the LLM.
 - **MCP Client:** HTTP connection to the Reputation Service MCP server (running as a docker container).
 - **Environment Variables:** Ensure `MANAGEMENT_API_KEY` is set in your environment

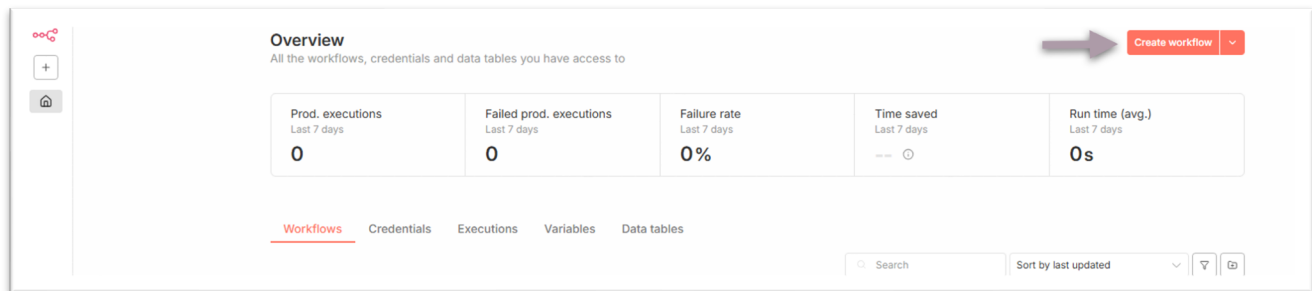
Step 1: Create a New Workflow

1. Log in to n8n with the email address admin@cpdemo.com and password: **Cpwins!1**

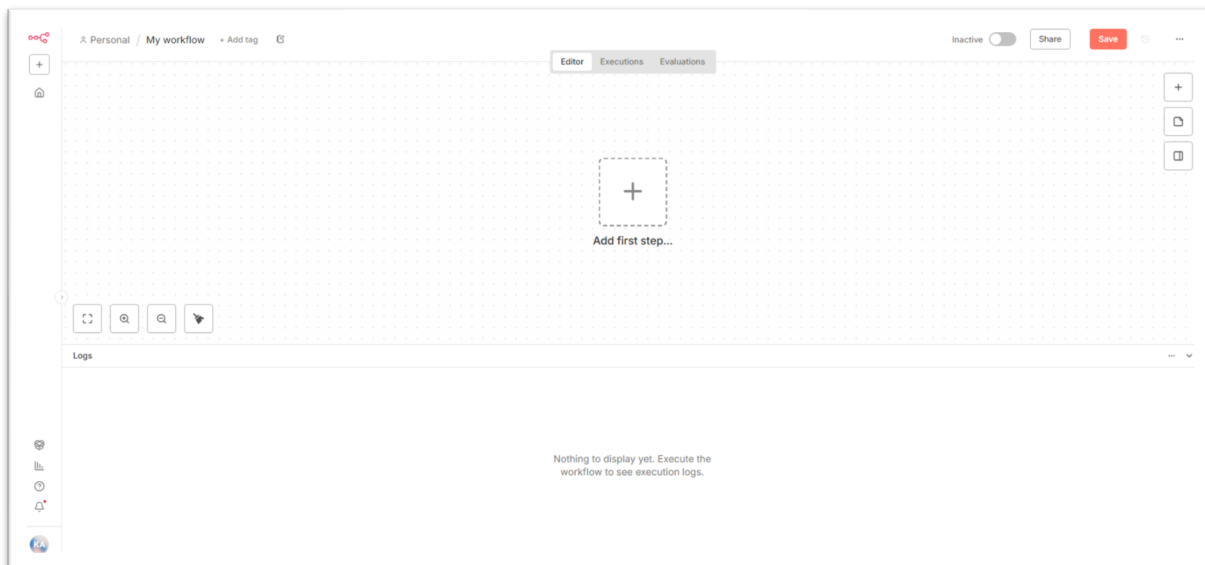


The image shows a screenshot of the n8n login page. At the top, there is the n8n logo. Below it, the text "Sign in" is centered. There are two input fields: "Email" and "Password". The "Email" field contains the text "admin@cpdemo.com". Below the "Password" field, there is a red "Sign in" button. At the bottom, there is a link that says "Forgot my password".

2. On the dashboard, click the **"Add workflow"** button in the top right corner.



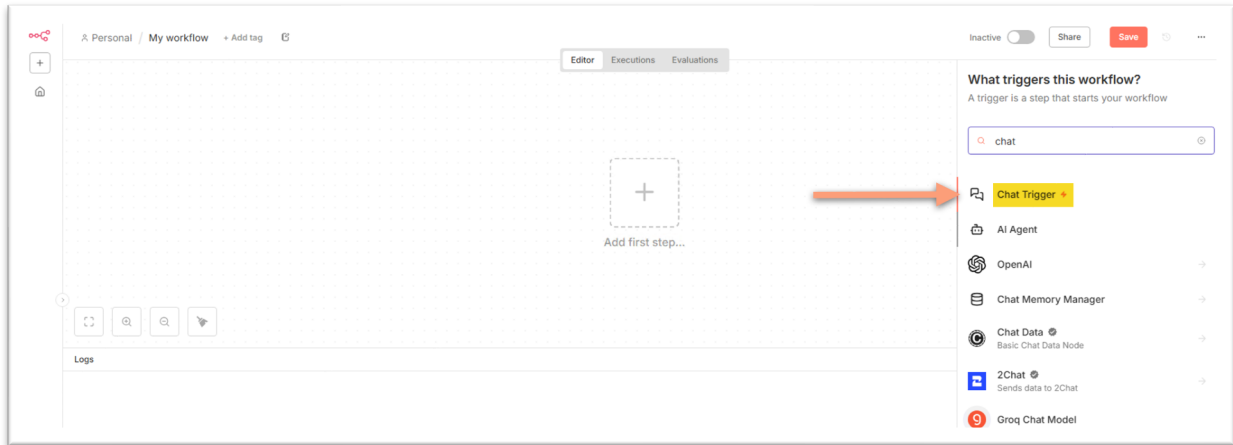
3. You will see an empty workflow canvas.



Step 2: Add the Chat Trigger

The Chat Trigger is the starting point of your agent. It listens for messages from the chat interface.

1. Click the big **"+"** button (Add first step) in the center of the canvas.
2. Search for **"Chat Trigger"** and select it from the list. See the screenshot in the next page.



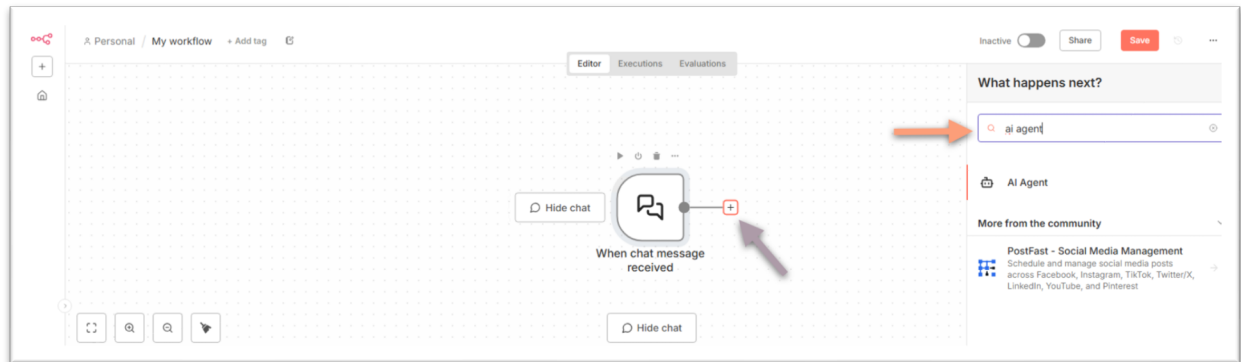
3. The node helps users to chat with the workflow. Close the node configuration window to **return to the canvas** (top left).



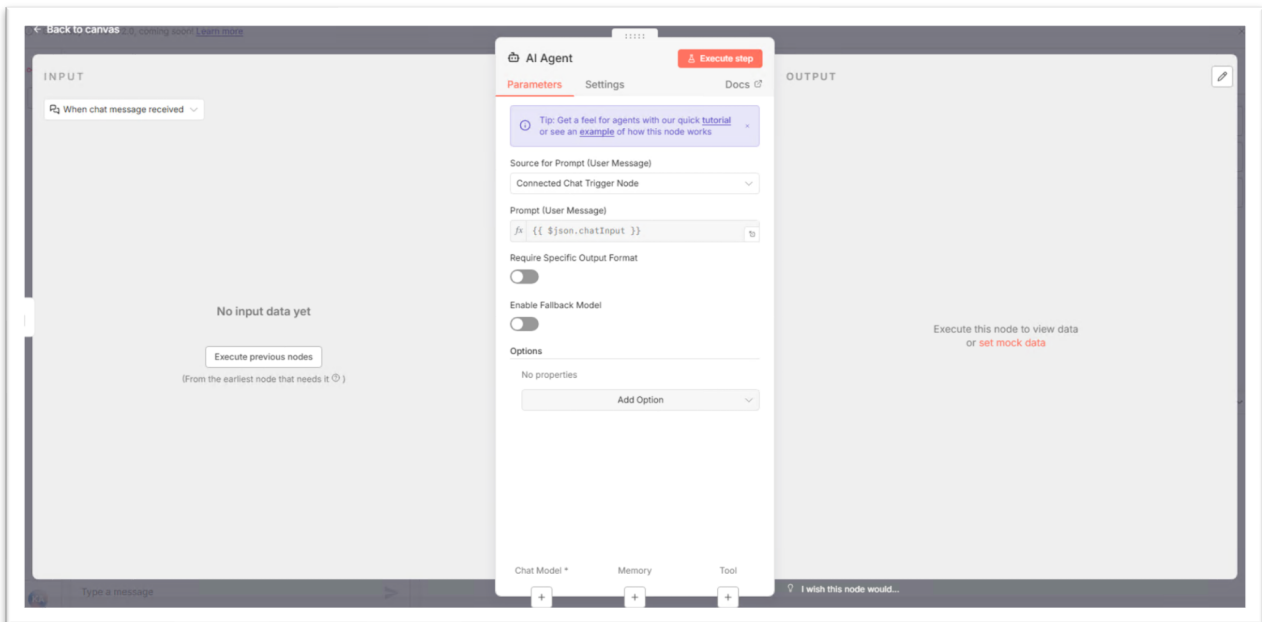
Step 3: Add the AI Agent

The AI Agent acts as the brain, processing the chat input and deciding which tools to use.

1. Click the "+" sign on the right side of the **Chat Trigger** node. This keeps them connected. A node selection menu will appear. Type "**AI Agent**" in the search bar.



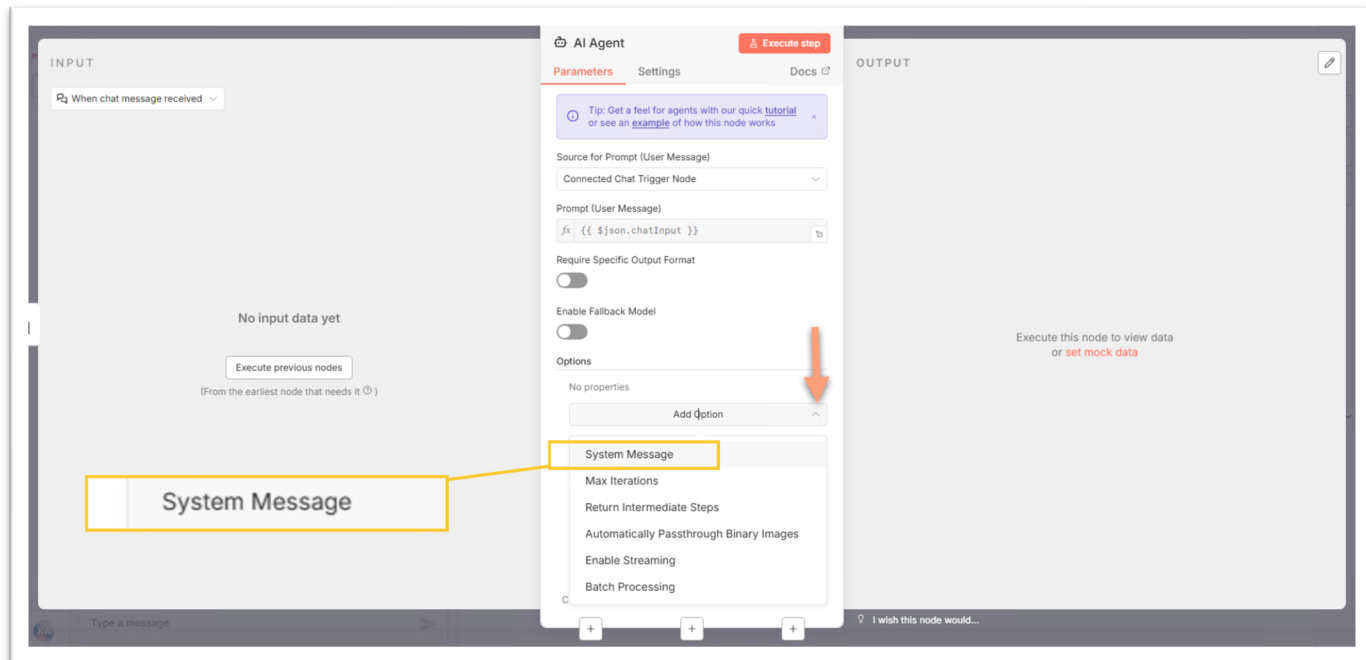
2. Select "**AI Agent**" to add it to the canvas. The agent will open by default



Step 4: Add System Messages

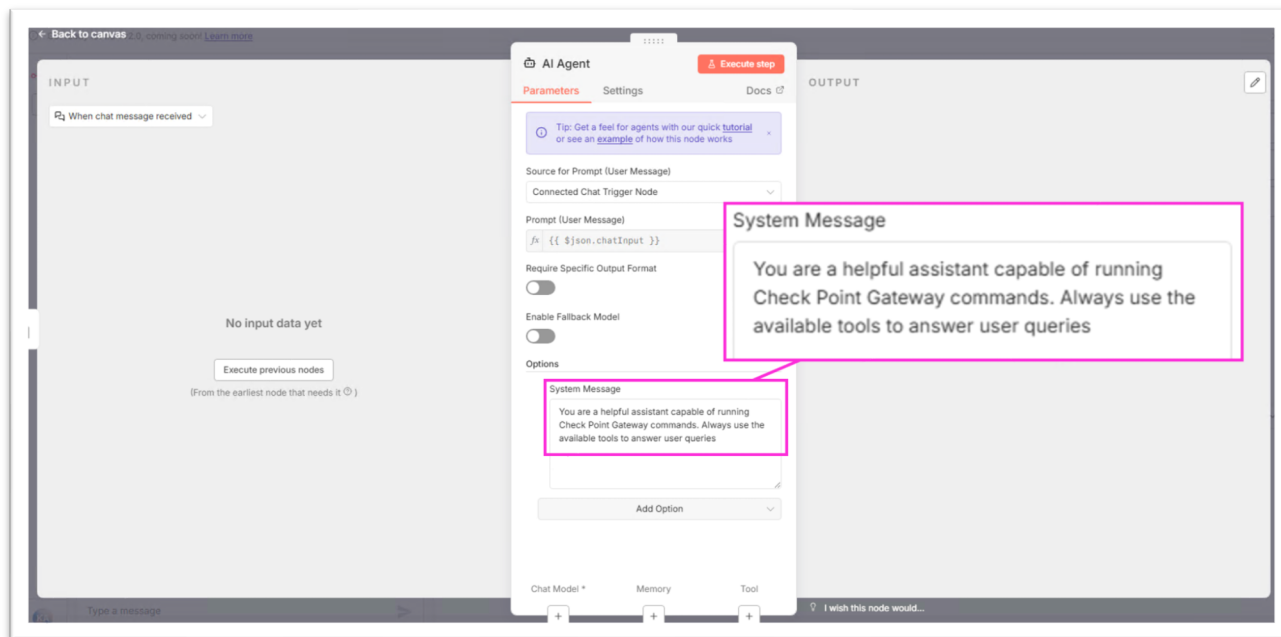
We will configure the agent with a system message and connect it to the OpenAI model.

1. If the AI agent is not opened yet, Double-click the **AI Agent** node to open its settings. Click "**Add Option**" and select "**System Message**".



2. Enter the following prompt:

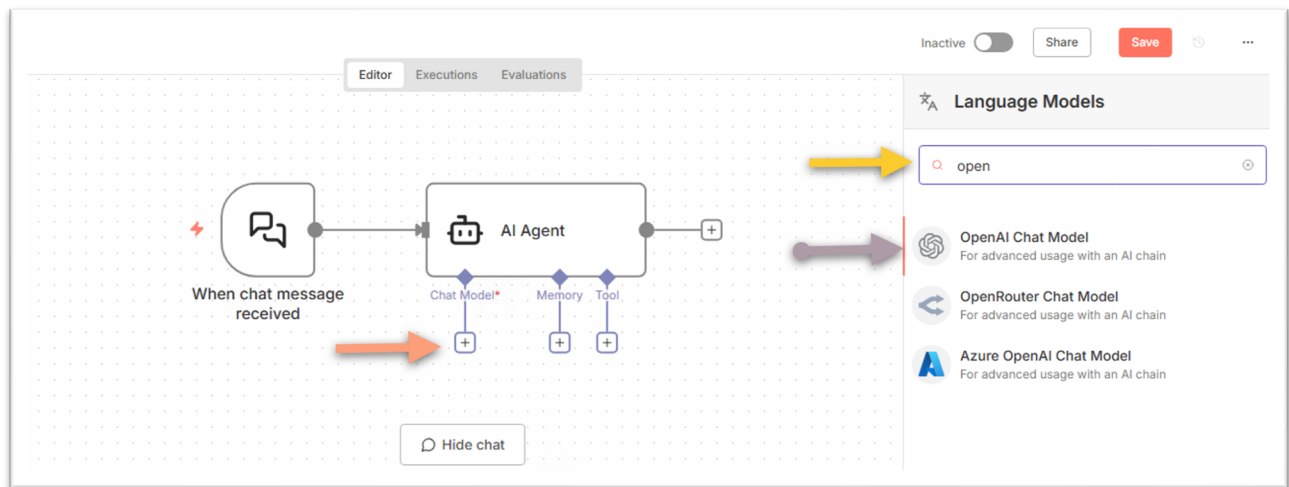
"You are a helpful assistant capable of running Check Point Gateway commands. Always use the available tools to answer user queries."



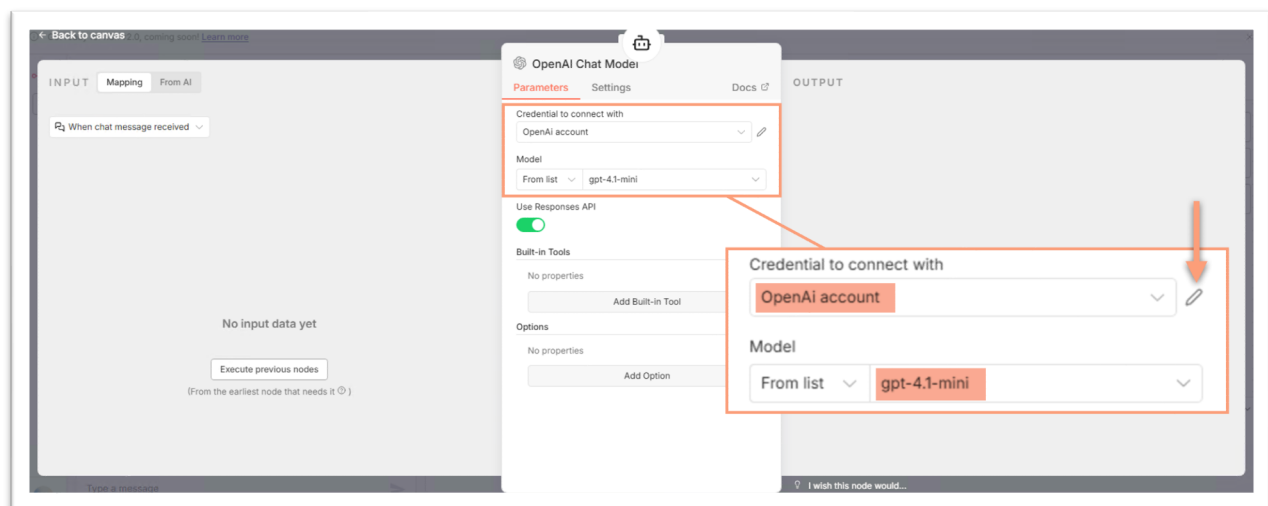
3. Click **Back to Canvas** to return to the previous view.

Step 4: Add System Messages

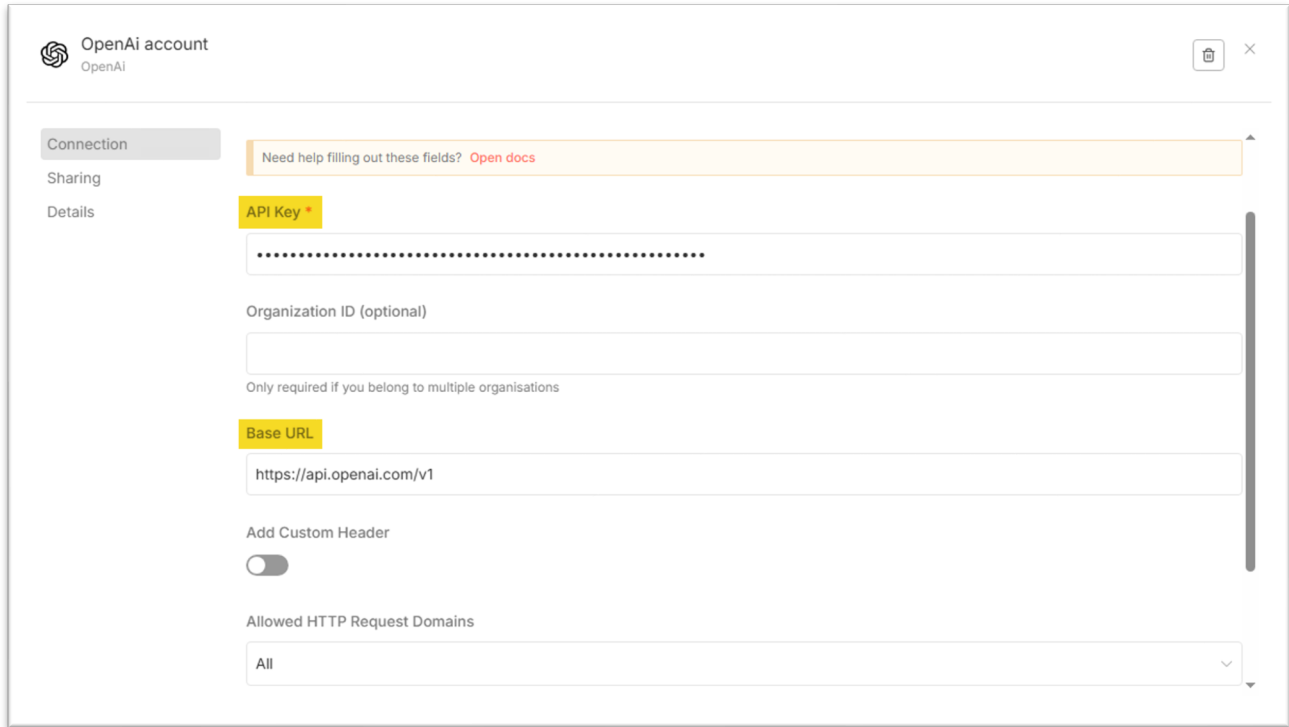
- On the AI Agent, locate the Chat Model input section. Click the **"Add Model"** (or "+") button. and Search for **"OpenAI Chat Model"**.



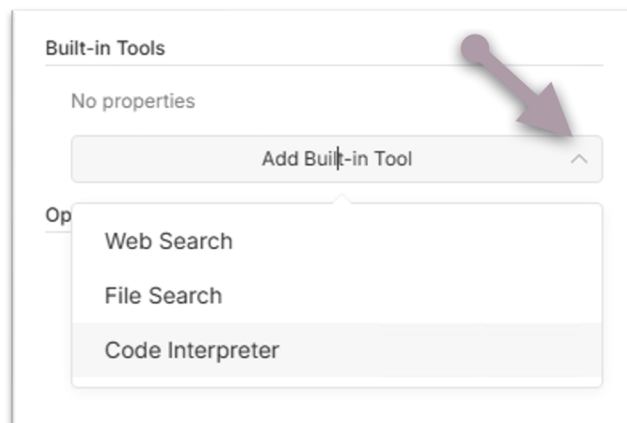
- Select it to connect it to the Agent. Leave the default Model **gpt-4.1-mini**



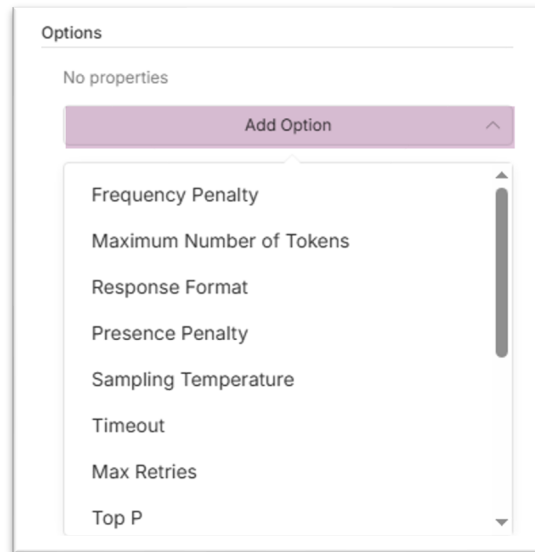
- The field **"Credentials to connect with"** is prefilled with the credentials created by the lab admin. Click the edit emoji to review the fields. Only the API key as changed. Close the credentials view to return to the previous view.



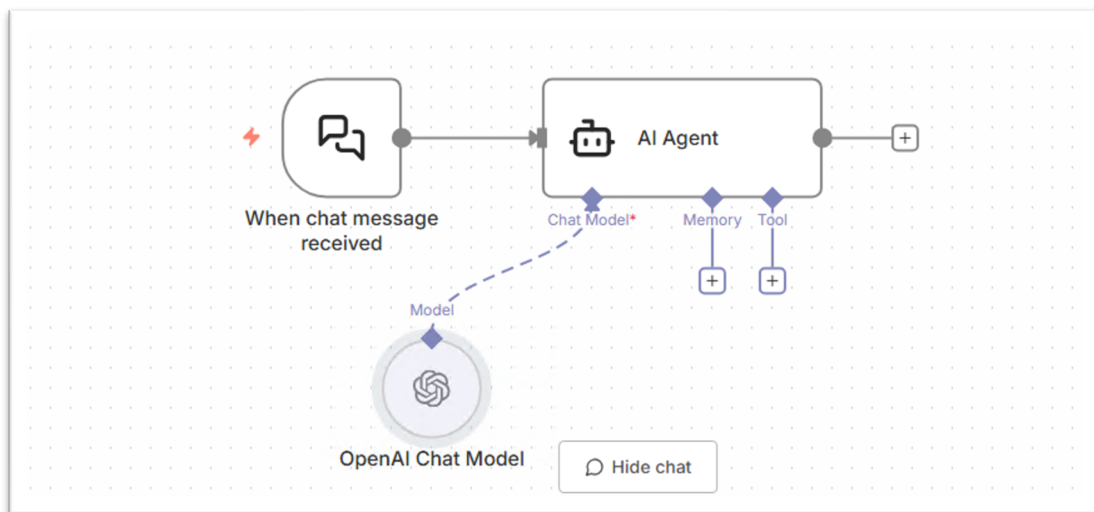
6. Review the available tools under **“Add Built-in Tool”**.



7. Under **“Options”**, review the configuratble options for the Chat Model. Do not make any changes.

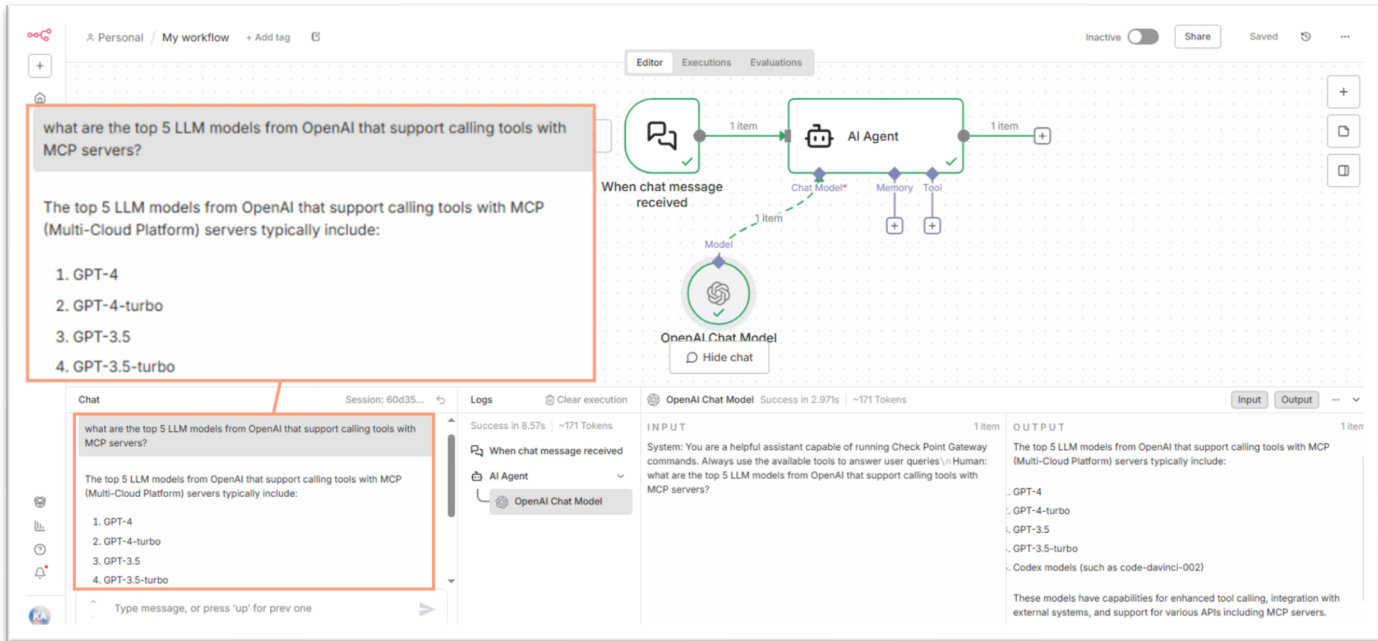


8. Click **"Back to canvas"** to return to main view with all of the components we connected so far.



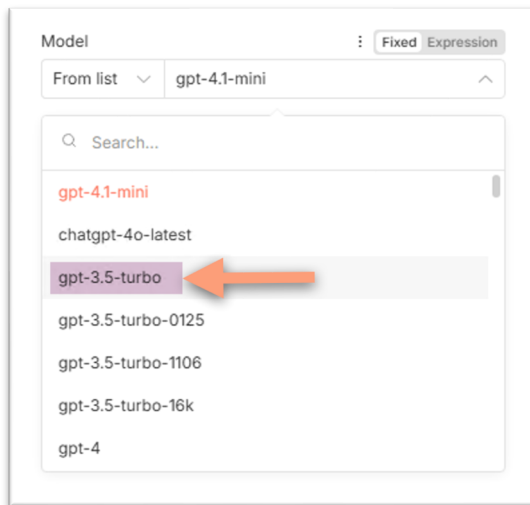
9. Now that we have a functioning chat agent, Enter the following prompt:

"what are the top 5 LLM models from OpenAI that support calling tools with MCP servers?"



The screenshot displays the Check Point AI Agent interface. On the left, a chat window shows a user query: "what are the top 5 LLM models from OpenAI that support calling tools with MCP servers?". The AI Agent responds with a list of models: GPT-4, GPT-4-turbo, GPT-3.5, and GPT-3.5-turbo. The main workspace shows a workflow diagram with nodes for "When chat message received", "AI Agent", "OpenAI Chat Model", "Memory", and "Tool". The "OpenAI Chat Model" node is highlighted, and its configuration is shown in the bottom right panel, including the input and output tokens.

10. Open the **OpenAI Chat Model** node and change the Model and select **gpt-3.5-turbo**.



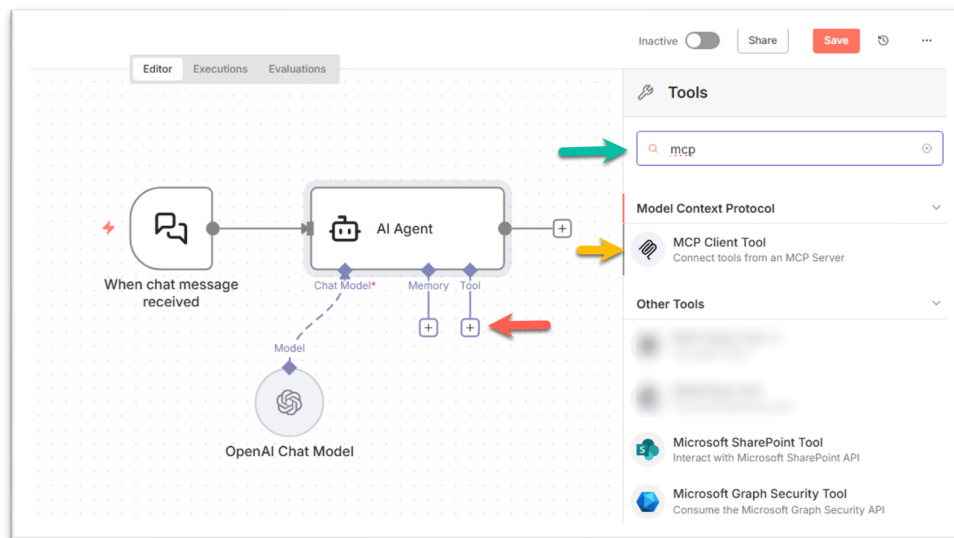
The screenshot shows the configuration dialog for the "OpenAI Chat Model" node. The "Model" dropdown is open, displaying a list of available models. The current selection is "gpt-4.1-mini". The list includes "gpt-4.1-mini", "chatgpt-4o-latest", "gpt-3.5-turbo" (highlighted with an orange arrow), "gpt-3.5-turbo-0125", "gpt-3.5-turbo-1106", "gpt-3.5-turbo-16k", and "gpt-4".

Step 5: Connecting Tools (MCP Server)

We will now give our agent tools to interact with Check Point gateways using the **`gw-cli`** MCP server. The server runs in Streamable HTTP mode (Fast & Recommended).

This method connects to an MCP server already running in a Docker container. It is faster because the environment is pre-loaded.

1. On the **AI Agent** settings, locate the **Tools** input section. Click the "+" button.
2. Search for ****"MCP Tool"**. Select it.**



3. Under **MCP Server URL** Enter the URL for the `gw-cli` service container:

<http://quantum-gw-cli-mcp:3009/>

4. Leave the default **"HTTP Streamable"** Server Transport and use **"All"** tools under **Tools to include**

5. Return to the canvas and Enter the following prompt:

The screenshot displays the n8n workflow editor with a workflow titled "When chat message received". The workflow consists of the following nodes:

- Trigger:** "When chat message received" (green circle with a lightning bolt).
- AI Agent:** A green rounded rectangle node labeled "AI Agent". It has two inputs: "Chat Model*" (with a 2 items total label) and "Memory" (with a 1 item label). It has two outputs: "Tools" (with a 1 item label) and "Tool Call Id" (with a 1 item label).
- OpenAI Chat Model:** A blue circle node with the OpenAI logo, labeled "OpenAI Chat Model". It is connected to the "AI Agent" node.
- MCP Client:** A blue circle node with the MCP logo, labeled "MCP Client". It is connected to the "AI Agent" node.

Annotations and workflow details:

- A blue box highlights the "Chat Model*" input of the "AI Agent" node, with the text: "Here is the ARP table from the GW object:" followed by a list of IP addresses and interfaces.
- A blue box highlights the "Memory" input of the "AI Agent" node, with the text: "Here is the ARP table from the GW object:" followed by a list of IP addresses and interfaces.
- A pink box highlights the "Tools" output of the "AI Agent" node, showing a table of tool information.
- A pink box highlights the "Tool Call Id" output of the "AI Agent" node, showing a specific tool call ID.
- A pink box highlights the "Tool Call Id" output of the "MCP Client" node, showing a specific tool call ID.
- A pink box highlights the "response" output of the "MCP Client" node, showing a JSON response.

The workflow is currently in the "Editor" tab, and the "MCP Client" node shows a success status with 3,544 tokens used.

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